Fossil Fuel Forest Fires

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Abstract

The purpose of this paper is to analyze and make inferences about how human activity has interacted and affected forest fires. With a devastating attack on the environment and its inhabitants, which includes humans, This is all due to the amount of fossil fuels that are being produced in factories and normal human day to day activity that could cause harm to animals and plants in a forest ecosystem. The paper will be describing the problem and how the problem originated from and its rise to dominate the woods. It will also go over the effects on the four spheres and two interactions from those spheres. Other spheres include the Lithosphere or Geosphere, Hydrosphere, Biosphere, and Atmosphere. The paper will also go over the importance of learning about this event and why it is a necessity to stop and prevent this at all costs.

Fossil Fuel Forest Fires

Throughout the Earth's history, forest fires have grown to a tremendous amount and the result of millions of trees being burned to the ground. One of the main reasons for this is the fossil fuels that are being released into the atmosphere and causing these forest fires to happen. As these fossil fuel companies expand and increase production, so will the harm caused by fossil fuels to all of the animals and plants that live in the forest habitat. The ecosystem in that area will be ruined. There are many areas to address such as the effects of the spheres which include the Lithosphere or Geosphere, Hydrosphere, Biosphere, and Atmosphere. This paper will analyze these things and explain how and why it happens and what humanity can do to stop it. Also, if an event causes an amplifying feedback or stabilizing feedback loop on the sphere.

Two Interactions and Event Lithosphere

The lithosphere is an important part of the ecosystem. Most fossil fuel production is man made and that can cause some forest fires while other forest fires can be natural. Fossil fuels contribute to global warming with how the carbon dioxide gets trapped in the atmosphere and can cause climate change. The hotter temperatures and drier air can cause ignition and forest fires. The heat from the forest fires can cause massive damage to the lithosphere. "The heat from the fire can cause the rocks and soil to expand and contract, which can lead to cracks and fissures in the lithosphere. These cracks can then act as conduits for water and other fluids to flow through, which can further damage the rocks and soil." (The damage a forest fire can do to the lithosphere, n.d.). There is even discoloration with the smoke and ash. The main cause of the wild fires is from the atmosphere being too dry. Soil composition is affected with erosion that can cause more harm to the lithosphere. " The ash and heat produced by a forest fire can have a positive impact on the Geosphere, which may result in beneficial soil conditions. This would benefit crops as well as people who would be able to generate revenue from them." (The damage a forest fire can do to the lithosphere, n.d.). The event causes an amplifying feedback to the lithosphere with the soil and earth being dried out and contributing to the fire. "It is a key indicator of fire danger and drought, Phillips said, and measures how much the air is sucking moisture out of soil and plants, which then ultimately become fuel for wildfires." (Ramirez, 2023).

Two Interactions and Event Hydrosphere

The hydrosphere has a very important role in our ecosystem. The hydrosphere gives all life on earth water to them to survive. All life is dependent on water. However, things like fossil fuels can cause many problems when it comes to the Hydrosphere. For one, fossil fuels can cause man-made/ man-influenced forest fires as the production of fossil fuels are man-made while the results of forest fires are a result of those actions. "84% of wildfires are due to human negligence. Unattended campfires, discarded cigarette butts, and improper burning of debris may all contribute to the creation of wildfires. Wildfires due to human error may be more common, but are likely to destroy less than a wildfire started by natural causes. Human made wildfires are often reported immediately and are able to be contained before excessive destruction has occurred." (Admin, 2021). As to how it affects the hydrosphere, the soil becomes dry due to the lack of vegetation and humid climate. Trees that take in some of the water are now gone and all that remains is ash. The rainfall comes down and causes many problems. "This water allows the transportation of harmful chemicals, various pollutants, and other damaging materials into streams, rivers, and large bodies of water. Homes destroyed in a fire risk the possibility of exposing dangerous materials like asbestos, and lead paint. Ultimately, these materials can be detrimental to our health and environment. They have the potential to upset the balance of water ecosystems, contain drinking water, and pollute essential nutrients and healthy resources." (Admin, 2021). The floods come in and destroy life by bringing in ashes to marine life which forces them to move. Purifying water is really expensive without trees doing some of the work for humanity. The event causes an amplifying feedback to the hydrosphere with the soil and earth being dried out and trees not being able to help filter out the ash and other pollutants in the water that cause life harm.

Two Interactions and Event Biosphere

The biosphere is very affected by the forest fires. The forest fires are man- made due to fossil fuels that cause it. The warmer temperatures allow more fires to spark due to the dryness in the air. "The most direct effect is usually on the biosphere, as the fire will often destroy vegetation and disturb wildlife habitats." (The damage a forest fire can do to the lithosphere, n.d.). The fire destroys the biosphere through destroying habitats and vegetation that is essential for plants and animals to thrive in. Animals have a difficult time escaping the fire and can cause them to become endangered to the location where they can live. "Damage to trees and plants not only affects the plant life, but it also affects the animals that rely on them to survive. Animals will struggle to find new, suitable habitats if they lose food and shelter." (The damage a forest fire can do to the lithosphere, n.d.). The event causes an amplifying feedback to the hydrosphere with the soil and earth being dried out and trees not being able to help filter out the ash and other pollutants in the water that cause life harm. ""The study takes what we know about the strong relationship between climate and burned area, and extends this understanding to the role of big fossil fuel emitters," (Ramirez, 2023). Fossil fuels production must stop as it is one of the biggest contributors to climate change. "Hotter, drier weather caused by climate change and poor land management create conditions favorable for more frequent, larger and higher-intensity wildfires." (Vizzuality, n.d.). The event causes an amplifying feedback to the biosphere with how the fire can impact the destruction of vegetation and how it brings lots of harm to animals and plants living in that area.

Two Interactions and Event Atmosphere

The atmosphere can contribute to forest fires. Fossil fuels are man-made while forest fires are both man-made and natural. With the atmosphere being too dry from fossil fuels, forest fires can build up and start more easily. It is the most common cause for forest fires. Along with the pollution caused by fossil fuels, "The fire may also release pollutants into the atmosphere, which can then have an effect on the climate." (The damage a forest fire can do to the lithosphere, n.d.). Fossil fuels also release carbon dioxide into the atmosphere which causes more heat to be trapped in the atmosphere and heat up the planet. Pollution is also being caused by both fires and fossil fuels as the flumes of the fires spread throughout and go into the atmosphere and thus, trapping even more heat. "Major oil and gas companies contribute roughly two-thirds of total industrial aerosol emissions, according to the study, which used fossil fuel emissions data through 2015, the latest available, and held figures constant from 2015 to 2021." (Ramirez, 2023). Forest fires produce a lot of greenhouse gasses like carbon dioxide and methane. "The plants that re-colonize burned areas remove carbon from the atmosphere, generally leading to a net neutral effect on climate. However, when fires burn more frequently and consume larger areas, as they are doing with climate change, the released greenhouse gasses may not be completely removed from the atmosphere if plants can't grow to maturity before burning, or if the plants that re-colonize are less efficient at carbon uptake." (*Wildfires and climate change,* 2023). The event causes an amplifying feedback to the atmosphere with how the fire can impact the destruction of vegetation and bring lots of harmful gasses to the atmosphere.

Conclusion

From learning about the effects that fossil fuels have on the four spheres in terms of forest fires, there is now proof and evidence showing how harmful they are for the planet and for life on Earth. There is a lot to see in terms of what harm they could cause and the fact that humanity is still using fossil fuels and causing these problems is outrageous. There are a lot of problems with this with how the biosphere is affected with how vegetation is dying and animals are forced to leave. The hydrosphere and its ability to pollute water sources. The atmosphere with it being polluted and causing the air we breathe to be contaminated. Even worse, it is causing more of the forest fires to happen and the cycle to continue. However, from analyzing and showing how harmful these things are. Scientists will soon find a solution to this and hopefully solve this issue.

References

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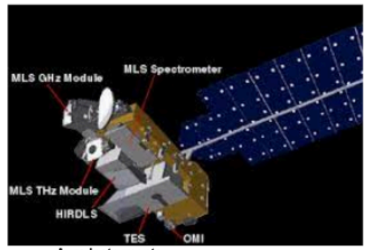
Aura Mission

Mission and Objectives

The mission is to observe the ozone layer and gather any data or information about it. They also monitor the tropospheric pollutants and use instruments to gather information about gasses in the atmosphere.



Aura Instruments

Issues and Problems/Overview

- Short term mission
- Insufficient power generation by the solar array
- Spacecraft is slowly drifting over time
- Minor drift
- Losing altitude

Progress/Key Dates/Timeline

- Aura spacecraft launched on July 15, 2004
- UARS made stratospheric constituent measurements from 1991-2005.
- 16 day repeat cycle, 233 revolutions per cycle

Feb 17, 2024.